MONDRIAN AND SAENREDAM probably knew a story recorded by Pliny the Elder, in his *Natural History, Book 35*, about two great artists of Ancient Greece, Apelles and Protogenes.

> A clever incident took place between Protogenes and Apelles. Protogenes lived at Rhodes, and Apelles made the voyage there from a desire to make himself acquainted with Protogenes's works, as that artist was hitherto only known to him by reputation.
>
> He went at once to his studio. The artist was not there but there was a panel of considerable size on the easel prepared for painting, which was in the charge of a single old woman.
>
> In answer to his enquiry, she told him that Protogenes was not at home, and asked who it was she should report as having wished to see him. 'Say it was this person,' said Apelles, and taking up a brush he painted in colour across the panel an extremely fine line;
>
> and when Protogenes returned the old woman showed him what had taken place. The story goes that the artist, after looking closely at the finish of this, said that the new arrival was Apelles, as so perfect a piece of work tallied with nobody else; and he himself, using another colour, drew a still finer line exactly on the top of the first one, and leaving the room told the attendant to show it to the visitor if he returned and add that this was the person he was in search of;
>
> and so it happened; for Apelles came back, and, ashamed to be beaten, cut the lines with another in a third colour, leaving no room for any further display of minute work.
>
> Hereupon Protogenes admitted he was defeated, and flew down to the harbour to look for the visitor; and he decided that the panel should be handed on to posterity as it was, to be admired as a marvel by everybody, but particularly by artists.

MODERN ART in the Western tradition. sounds like it began with the lines of Apelles and Protogenes, long before Saenredam and Mondrian.

READING

- Michel Foucault. The order of things. In *Order of Things, The*. Taylor and Francis Group, United Kingdom, 2005. ISBN 9780415267366 Download paper

WRITING

Foucault's essay is challenging, but rewarding if you spend time with it. His was the first important paper on *Las Meninas* and the puzzle of its "representation of representation".

After reading Foucault, respond to "representation of representation" in another painting of your choice, like Vermeer's *Art of Painting* or Weyden's *St. Luke*. What do these paintings say about the nature of art or the self-image of the artist themselves?

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- Svetlana Alpers. *The Art of Describing*. University of Chicago Press, 1983. ISBN 978-0-226-01513-2
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- Joseph Leo Koerner. *Dürer's Mobility*. National Gallery Global, London, 2022
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Updated: April 7, 2025

Week Six - The Art of Describing

Thursday, 6 March 2025, 12:45 - 2:15 PM EST.
M-Lab, Harvard Art Museums

The Netherlands in the time of Rembrandt and Vermeer was also a time of Dutch scientific achievement. A new type of science involved *seeing* and *describing* the natural world. Galileo Galilei (1564-1642) was one leader who used telescopes to see and describe the night sky. But being in Florence, Galileo was an exception. Most contemporary scientists who found new ways to use optics to see and describe the world were in Northern Europe. The Danish Tycho Brahe (1546-1601) built the largest observatory in Europe. The German Johannes Kepler (1571-1630) improved Galileo's telescope and mathematical understanding of the solar system. The Dutch Antony von Leeuwenhoek (1632-1723) developed a microscope that enabled him to discover microorganisms and bacterial life.

When a new world is seen – whether with telescope or microscope – it has to be described. Quantitative and reliable visual description – *pictorial art* – became necessary for scientific communication, as important as the written word or equation.

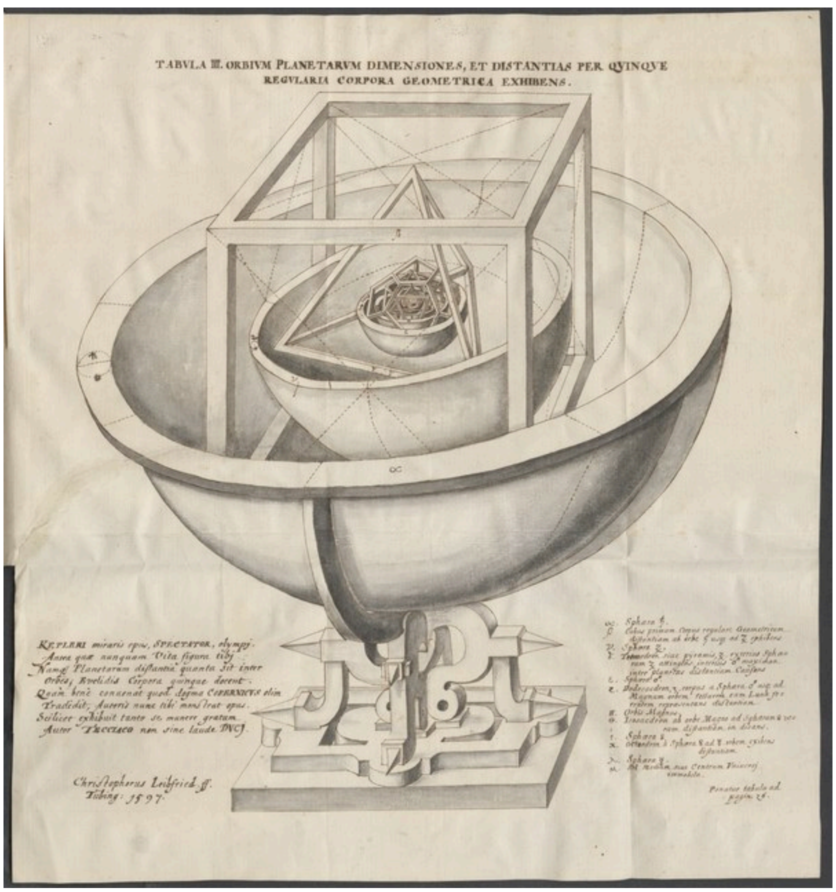


Figure 134: Platonic solid model of the Solar System from *Mysterium Cosmographicum* by Kepler.

NORTHERN EUROPE was especially receptive to optical science. The writings of Sir Constantijn Huygens, Lord of Zuilichem, (1596-1687), provides one window into contemporary Dutch attitudes (Fig. 135). Constantijn was a polymath. He corresponded with Descartes, championed Rembrandt, and inspired his son to become a mathematical physicist . Christiaan Huygens (1629-1695) would discover 'Huygens Principle' of wave propagation.

Figure 135: *Portrait of Constantijn Huygens and his Clerk* by Thomas de Keyser. Huygens's table is the story of his duties, interests, and talents – a musical instrument, architectural plans, and terrestrial and celestial globes. Link to painting at the National Gallery in London

Constantijn had the opportunity to use a camera obscura that was bought by his friend Cornelis Drebbel, another leading scientist. His reaction reveals the impact of this new image-making device and the importance of visual over written communication.

> It is not possible to describe for you the beauty of [the camera obscura] in words: all painting is dead by comparison, for here is life itself, or something more noble, if only it did not lack words.

A POEM BY CONSTANTIJN is similarly revealing. Seeing is both a means to knowledge and way to experience God's creation:

> O you who give the eyes and the power,
> Give eyes through this power:
> Eyes once made watchful, Which see the totality of all there is to see.

LENSES AND MIRRORS in the 17th century were not perfect. Images were distorted. The circumspect worried whether optics revealed truth or lies. Optical devices were validated in 1604 when Kepler discovered that the human eye literally *works* like a camera obscura. The eye, like the camera obscura, is equipped with lens and pinhole that projects images onto the back surface. Kepler wrote: "ut pictura, ita vision" or "sight is like a picture".

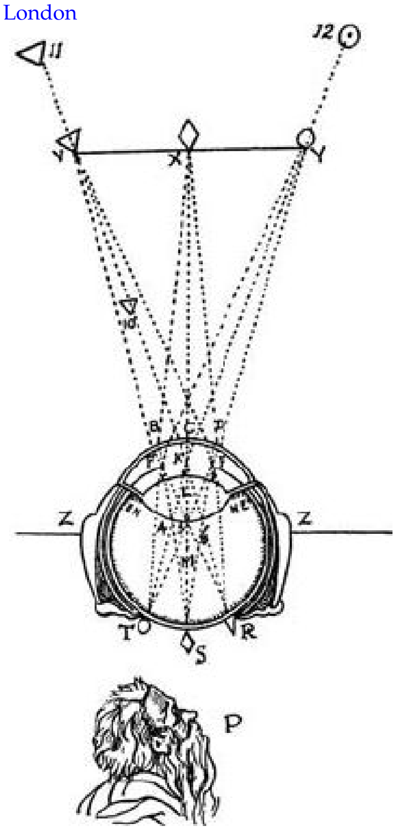
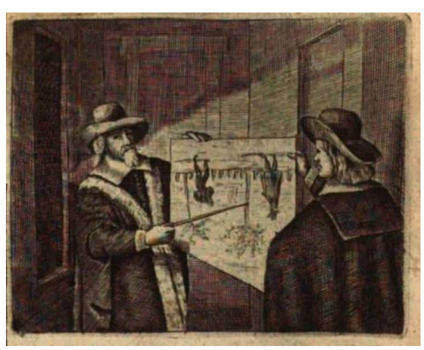


Figure 136: Illustration of the theory of the retinal image from *La Dioptrique* by Descartes.

IN A DUTCH HANDBOOK BY VAN BEVERWYCK, *Wercken der Genees-Konste* from 1667, two men illustrate the operation of the eye by standing in a dark room with a pinhole fitted by a lens. The outside world is cast onto a surface, just like an image is cast onto the back of the eye. In the camera obscura, pictures are painted with rays of light, not with brush and pigment.



NORTHERN EUROPEAN ART was long known for detailed, precise, and patiently crafted descriptions of the natural world. When Constantijn looked into Drebbel's microscope, he found another visual worlds that needed to be described. These descriptions required new art.

CONSTANTIJN HUYGENS became a supporter of Leeuwenhoek (1632-1723), a contemporary of Vermeer (1632-1675) in the small town of Delft. Leeuwenhoek discovered microorganism by inventing the most powerful single-lens microscope in Europe. His microscope had magnification and clarity far beyond what Drebbel would had shown to Constantijn. Leeuwenhoek made the first drawings of bacteria, which he called *animalcules.*

In the 17th century, learning to draw was as essential to a complete education as learning languages or mathematics. As a student, Constantijn had wanted to study drawing with the Dutch realist Jacob de Gheyn II (1565-1629). De Gheyn's representations of flora and fauna were famed for scientific precision (Fig. 138). Such talents were needed for the new world that Constantijn and Leeuwenhoek saw in the microscope. Constantijn wrote:

> For in fact, this concerns a new theater of nature, another world, and if our revered predecessor De Gheyn had been allotted a longer life span, I believe he would have advanced to the point to which I have begun to push people (not against their will); namely to portray the most minute objects and insects with a finer pencil, and then to compile these drawings into a book to be given the title of the *New World* .

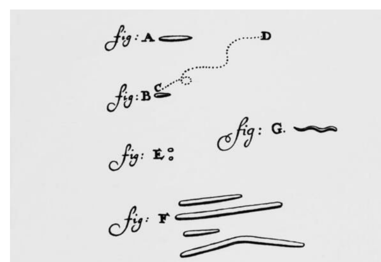


Figure 137: Animalcules from a letter by Leeuwenhoek from 1683

Figure 138: *Studies of a Fantastic Bird, Toad, Frog, and Dragonfly* by de Gheyn, 1596-1602. Link to Morgan Library and Museum.

FRANCIS BACON (1561-1626) – philosopher at Cambridge University and the Court of Elizabeth I – has been called the inventor of the modern scientific method. He advocated *empiricism* – knowledge by inductive reasoning and carefully seeing and describing Nature. Constantijn Huygens was influenced by Bacon, as were Robert Hooke (1635-1703) and Isaac Newton (1643-1727).

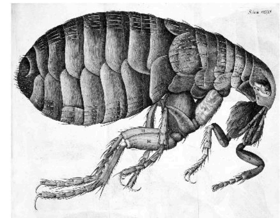


Figure 139: *Flea* from *Micrographia* by Robert Hooke, 1665

Robert Hooke corresponded with Leeuwenhoek and made his own microscopes. Huygens had advocated a scientific program of accurate visual representation of the microscopic world. This program was realized by Hooke when he published *Micrographia* in 1665. [8] Hooke used his microscopes to discover and describe the surprisingly exquisite architectures of insects.

[8] Martin Kemp. Hooke's housefly. *Nature*, 393(6687):745–745, 1998. ISSN 0028-0836

MICROGRAPHIA uses both words and images to describe Nature. Comparing Hooke's words with drawings reveals the weakness of language in describing the new worlds he saw in the microscope.

> The Eye of a Fly in one kind of light appears almost like a lattice, drill'd through with abundance of small holes... in the Sunshine they look like a surface cover'd with golden Nails' in another posture, like a surface cover'd with pyramids; in another with Cones; and in other postures of quite other shapes.

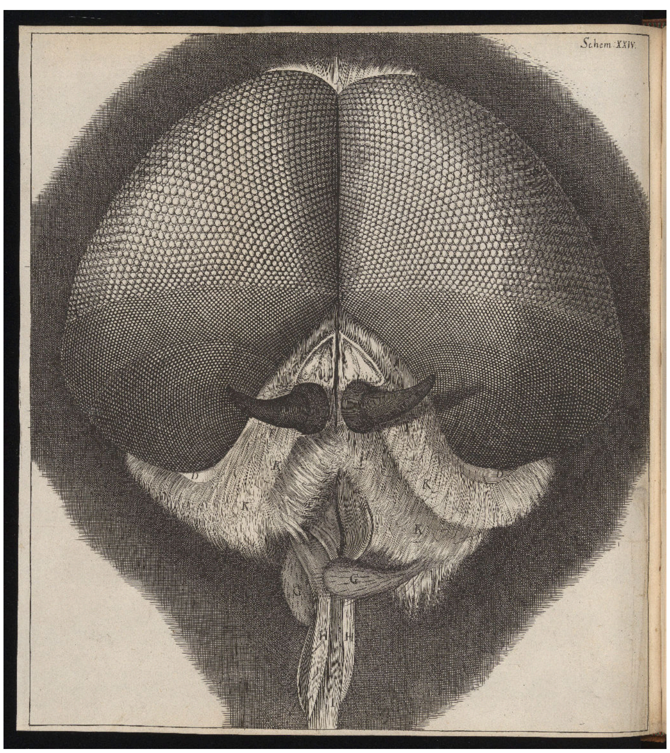


Figure 140: *Eyes and head of a grey drone fly* by Robert Hooke, 1665. Link to Wellcome Collection.

COMMON THEMES IN ART AND SCIENCE reveal shared cultural attitudes and tastes. When de Gheyn painted a sick mouse, he captured the details of its swollen eyes and fur standing on end from multiple angles. These drawings reflect the artist's devotion to photographic realism, but also a scientific process of thought as the artist contemplates the mouse, rolling it over in his mind.

Figure 141: *Four studies of a diseased mouse* by Jacques de Gheyn. Link to Rijksmuseum.

THE DUTCH STILL LIFE also reveals a taste for microscopic dissection and displaying every facet of an object. Specialists in the still life, like Pieter Claesz (1597-1661) and Willem Kalf (1619-1693), would paint tableau of cheese, fish, fruit, and serving vessels, revealing inside, outside, and underside. Foods are sliced open. Lemon rinds are unwound. Objects are dissected and exposed. The eye is drawn to every optical detail that is faithfully captured with paint and brush.

Figure 142: *Still Life* by Pieter Claesz, 1627. Still life did not develop into its own distinct style until the 17th century. Pieter Claesz in Haarlem specialized in these still lifes, devising different arrangements to display all dimensions of common objects. Link to Timken Museum.

ABRAHAM VAN BEYEREN (1620-1690) was another Dutch Baroque painter of still lifes. He often includes lemons, maximizing their view with sliced and unwound peel (Fig. 143). The lemon is deliberately exposed for the curious eye, not as it might sit in a fruit bowl. Here, the artist also includes himself reflected in the convex mirror of the silver jug.

HIGH-END ARTISTS often included expensive and rare collectibles in still life, a way of making paintings more attractive to the wealthy (Fig. 144). In the Netherlands, paintings were rarely commissioned and had to compete in a fierce market. Between 1640 and 1660, Dutch artists made *1.3 million paintings.* Artists pushed themselves to compete.

Figure 143: *Ostentatious still life with self-portrait of the artist in the silver jug* by Abraham van Beyeren, ca. 1655 Link to Mauritshuis .

Figure 144: *Still Life with Tazza* by Pieter Claesz, 1636. Various objects strewn on a table, including sliced lemon. The tazza is the lavishly decorated silver drinking vessel. Link to Mauritshuis

David Hockney came to alternative optical conclusion with a still life by Juan van der Hamen y León (Fig. 147). [9] Here, grapes, sliced melon, and sliced pomegranates are painted separately. None would stay fresh by the time that another was painted. This work looks like a montage of separately painted objects, each captured in a snapshot moment of time much briefer than painting requires. An optical tool would have let the artist work quickly, before the fruit rotted.

[9] David Hockney. *Secret Knowledge*. Viking Studio, New York, 2006. ISBN 978-0-14-200512-5

Figure 145: *Still Life with Fruit and Glassware* by Juan van der Hamen y León, 1626. The Museum of Fine Arts, Houston

KEPLER first described the retinal image as a “picture”. What the eye sees is effectively painted by rays of light on the back of the eye. This model of the eye has an underlying statement about visual truth and the validity of the camera obscura for depicting the world. Kepler likely owned a “tent camera obscura”, which he used for landscapes. The image formed by such a tent camera obscura would look like an image made by a modern camera.

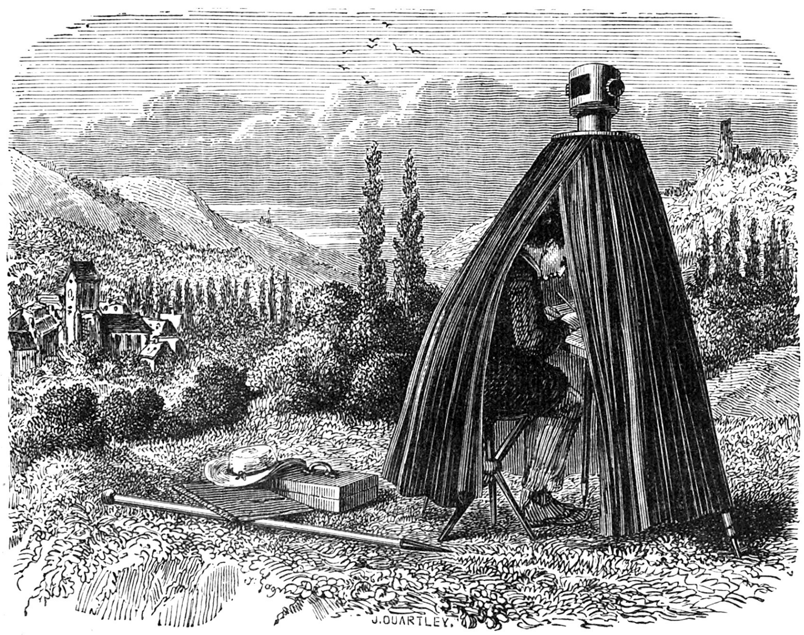


Figure 146: *Tent Camera Obscura* from an 1858 book on physics.

VERMEER'S *View of Delft* has the uncanny look of a color photograph. Vermeer had found a new approach to landscape painting. Delft has no obvious geometric design or narrative center. Instead, Vermeer captured a glimpse of a quiet city on a quiet morning. How did Vermeer see Delft in a way that looks like a color photograph?

A CAMERA OBSCURA might have helped Vermeer to see differently. Vermeer might have admired the unique way that a camera obscura created an instant and complete glimpse of the world. He then might have tried to replicate its visual effects, the same effects that occur in modern color photography, in his own painting.

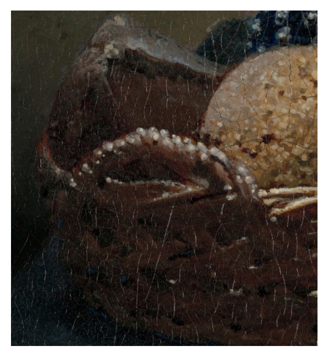
Figure 147: *View of Delft* by Vermeer
Link to painting.

Vermeer's *Milkmaid* also has the narrative effect of color photography. Vermeer captures a glimpse of a domestic scene, a quiet moment as the milkmaid pours. Linear perspective does not create a three-dimensional stage for a drama. Sunlight illuminates an everyday scene from life, which Vermeer transfers to paint.

Certain optical effects appear in Vermeer paintings that are unique to lens-based device. In both *The Milkmaid* and *The View of Delft*, Vermeer dabs 'circles of confusion', bright globules of coalesced highlights that simulate an out-of-focus effect caused by diffracting and reflecting light. Some circles of confusion are out of place, like from the non-reflective matte surface of the bread basket in *The Milkmaid*. Vermeer does not faithfully copy what he sees. He sprinkles optical effects from a camera obscura like ornaments.

Another optical effect in *The Milkmaid* is optical focus. Objects have sharp focus in an image plane and soft focus in other planes. Optical focus only happens when an image is made with a lens. The naked eye automatically and unconsciously focuses to every point in the real world. You cannot see variable focus unless you look at a flat image that was made by a fixed lens. Vermeer might have liked the focus of the camera obscura, and mimicked its effect in a painting.

So long as that woman from the
  Rijksmuseum
in painted quiet and concentration
keeps pouring milk day after day
from the pitcher to the bowl
the World hasn't earned
the world's end.
  –*Wislawa Symborska*



Another View of Delft is by Carel Fabritius (1622-1654). Fabritius has been called the most brilliant of Rembrandt's students. Ten years older and living in Delft, he might have taught Vermeer. Little remains of his work. He was killed by the Delft gunpowder explosion on 12 October 1654 that destroyed his studio and paintings. Only a dozen paintings survive, including *The Goldfinch* and *View of Delft*.

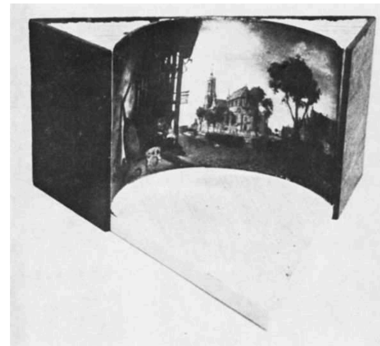


Figure 148: Interior view of a reconstruction of a perspective box that might have contained Fabritius's *View of Delft*.

Fabritius's *View of Delft* has a wide-angle perspective. It resembles the panoramic photograph when a smartphone is swept across a field of view. Now, Fabritius's painting is hung and seen on a flat surface in London. But the painting was probably originally mounted on a curved surface in a rounded box.

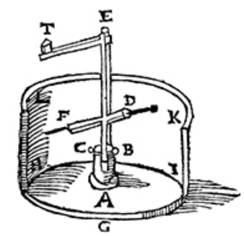


Figure 150: Baldassare Lanci device to project a view on the inside of a cylinder (Kemp, 1990)

Maybe the *View of Delft* is an act of imagination. The artist predicted, in his mind's eye, what a future smartphone might do. Another possibility is that Fabritius used a device, like a camera obscura with a curved surface. Such machines that mapped scenes onto cylindrical surfaces did exist (Fig. 150). A third possibility is that Fabritius copied what he saw through a lens. A wide-angle camera shot of the same scene in modern Delft has a similar optical distortion as Fabritius's painting (Fig. 149).

Figure 149: View of the church painted by Fabritius, but photographer through a double convex lens. From Wheelock (1973).

Figure 151: *The View of Delft* by Fabritius, 1652. This very small painting (15x30 cm) is a view of the Nieuwe Kerk in Delft. Perspective is emphasized by two musical instruments, typical virtuoso subjects for foreshortening. The violin recedes into picture space, as if very close to the viewer. The church loom forward, and the street bends as if curved by a lens. Link to National Gallery.

DELFT WAS DIFFERENT from what Fabritius saw in 1652 or Vermeer saw in 1661. In 1654, Delft was shattered by a gunpowder explosion, equivalent to 22 tons of TNT. This explosion destroyed a quarter of the city and killed hundreds including Fabritius. The explosion and its aftermath were painted by Egbert van der Poel. Vermeer's *View of Delft* cannot be a faithful visual image. In painting after painting, Vermeer leaves out grim reality in a city afflicted by war and plague. Vermeer chooses stillness. He chooses quietude.

Figure 152: *A View of Delft after the Explosion of 1654* by Egbert van der Poel, 1654, Link to National Gallery.

Figure 153: *A View of Delft During the Explosion of 1654* by Egbert van der Poel, 1654, Link to Rijksmuseum.

THE CAMERA OBSCURA became a scientific and commercial device in the 17 th century. But its basic optical principle, what happens when an image is cast by a pinhole in a darkened room, had been known for centuries. Before Kepler, Leonardo da Vinci had wondered whether the camera obscura was a model for the human eye. Da Vinci wanted to understand visual truth. He believed that the painter should depict what is projected into the eye:

> The painter's mind should be like a mirror, which transforms itself into the color of the thing that it has as its object, and is filled with as many likenesses as there are things placed before it. Therefore, painter, knowing that you cannot be good, if you are not a versatile master in reproducing through your art all the kinds of forms that nature produces – which you will not know how to do if you do not see and represent them in your mind.

THE FLOWERS, ROCKS, AND LANDSCAPE of his *Madonna of the Rocks* are painted with photographic realism, as da Vinci follows his own advice. Unlike Vermeer, da Vinci also takes liberties with what he sees, assembling objects to follow a painterly design. In *View of Delft* and other paintings, the painter and any painterly design seems to disappear. The image tells the whole story.

Figure 154: *Madonna of the Rocks* by Leonardo da Vinci, 1483-1486

Writing

Ernst Gombrich is the best known art historian of the 20 th century. His *The Story of Art* is the foundation of innumerable courses in Art History. Gombrich also taught Svetlana Alpers. Svetlana Alpers wrote her landmark *The Art of Describing*, about the special optical qualities of Northern European painting.

Download and read the Introduction to *The Art of Describing* (more if you want) and Gombrich's review of her book.

Comment on Alpers, and whether you think Gombrich agrees with her.

Painting

- Nard Kwast, Dutch reality TV star known for his recreation of Rembrandt's *Nightwatch* in the Rijksmuseum will lead a workshop to paint a Dutch Still Life. Link to slides

Reading

- Introduction of *The Art of Describing* Download entire book
- Review of *The Art of Describing* Download paper

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Week Seven - Art and Illusion

Thursday, 13 March 2025, 12:45 - 2:15 PM EST.
M-Lab, Harvard Art Museums

ART IS ILLUSION. A painter of realism is trying to create a flat picture that looks like a real three-dimensional scene. Attempts to use linear perspective or capture every optical details are, in part, attempts toward virtual reality. The painter is trying to create the illusion of seeing something real.

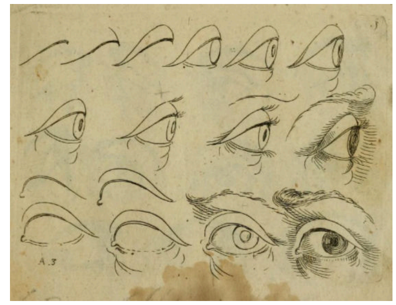

Figure 155: Odoardo Fialetti (1573-1638) published a drawing manual that describes how to draw the human eye as little more than successive outlines of its component parts.

Any painting can only partly succeed as an illusion. A flat image cannot reproduce the full multi-dimensional effect of seeing reality. Paintings can only evoke, in the mind of the viewer, similar mental images of seeing real objects. This fundamental connection between art and illusion has long been discussed. Pliny understood why fooling the eye with an optical illusion is not the same as fooling the mind of the viewer:

> the mind is the real instrument of sight and observation, the eyes act as a sort of vessel receiving and transmitting the visible portion of the consciousness.

The human brain transforms patterns of illumination – what happens when light is projected into the eye and onto the retina – into what we see.

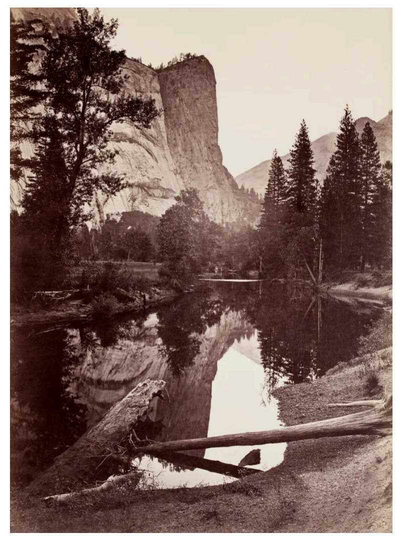

Figure 156: The shadows, reflections, landscape, sky, trees are easily recognized in this famous landscape photograph of Yosemite made by Carleton Watkins (1829-1916). Watkins was famous for landscape photography before Ansel Adams was born. He couldn't buy a simple film camera, and prepared his own film by hand in the field, pouring chemicals onto glass, exposing, and processing each unique photograph in the field. Watkins made these photographs with a room-sized camera obscura.

EVERY DRAWING IS AN ILLUSION. If an artist wants to picture an object, they often start by drawing its outline – tracing a line along the object boundary. But real objects are not bounded by any real line. Our brain creates lined boundaries around objects by neural processing of visual information. We see edges when different amounts of light are reflected by an object and by its surroundings. The brain effectively draws an outline around every object where the eye sees a sudden change in luminosity in adjacent regions of space. This is why outlining an object in a drawing creates the same mental image, an illusion, of the real object.

WE SEE OBJECTS BY SEEING EDGES. Our brain transforms edges in luminosity into outlines of different objects. This is why black and white photography works. There is no color in a black and white photograph. We see the shape and position of distinct objects only using edge detection based on sudden changes in light intensity. Within each object, the pattern of light intensity is a smoothly varying field. At the edge of every object, the brain uses sharp changes in light intensity to draw its boundaries.

THE RETINA AND BRAIN are hard-wired to detect the edges of objects on the basis of sudden changes in light intensity. Santiago Ramón y Cajal (1852-1934), an artist and neuroscientist, was the first to dissect and map the wiring of the retina. In skilled drawings of what he saw through his microscope, he describes extensive lateral wiring between adjacent areas of the retina viewed in cross-section.

We now know that lateral wiring is for edge detection. When a small circle of light is projected onto the retina, neurons inside the circle will be activated, but lateral wiring will *inhibit* nearby neurons just outside the bright circle.

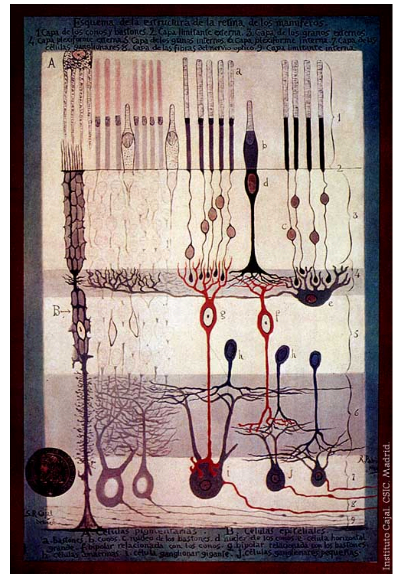

Figure 157: *Structure of the Mammalian Retina* c.1900 By Santiago Ramon y Cajal. a. Rods. b. Cones. c. Rod nucleus. d. Cone Nucleus. e. Large horizontal cell f. Cone-associated bipolar cell. g. Rod-associated bipolar cell. h. Amacrine cells. i. Giant ganglion cell. j. Small ganglion cells.

CENTER-SURROUND INHIBITION – lateral wiring – converts a two-dimensional image on the retina into a mental image with lined boundaries around each object. In computer vision, image processing filters work the same way, calculating spatial differences in light intensity to draw lines around objects.

Because the brain is wired to draw the outlines of objects in visual images, the brain also becomes good at identifying objects on the basis of simple boundaries. This is why we easily interpret the drawn line. This is why we easily recognize silhouettes, like the portrait of Thomas Jefferson by Raphaelle Peale (1774-1825).

Figure 159: An edge filter in computer vision that is based on lateral inhibition like the retina. From Wikipedia.

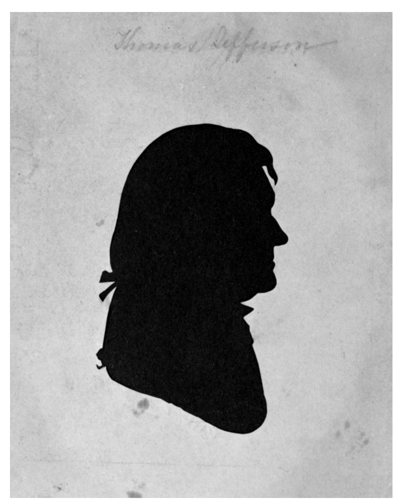

Figure 158: *Silhouette of Thomas Jefferson* by Raphaelle Peale, 1804.

CENTER-SURROUND INHIBITION helps us to see object outlines. But center-surround inhibition also makes us susceptible to an optical illusion when visual information spreads across sharp boundaries. This is why 'Mach bands' appear at any adjacency between areas with different uniform brightness. Next to the sharp boundary between dark and light areas, a band in the dark area will falsely appear darker and a band in the light area will falsely appear lighter.

Figure 160: Along the boundary between adjacent shades of grey in the Mach bands illusion, lateral inhibition makes the darker area falsely appear even darker and the lighter area falsely appear even lighter. From Wikipedia.

JESSE STILLWELL, a modern abstract painter, uses Mach bands on purpose. Stillwell uses the effect of center-surround inhibition in the retina to create illusions of color gradients from flat hues. His paintings work because *we infer what we see* from what the eye tells the brain.

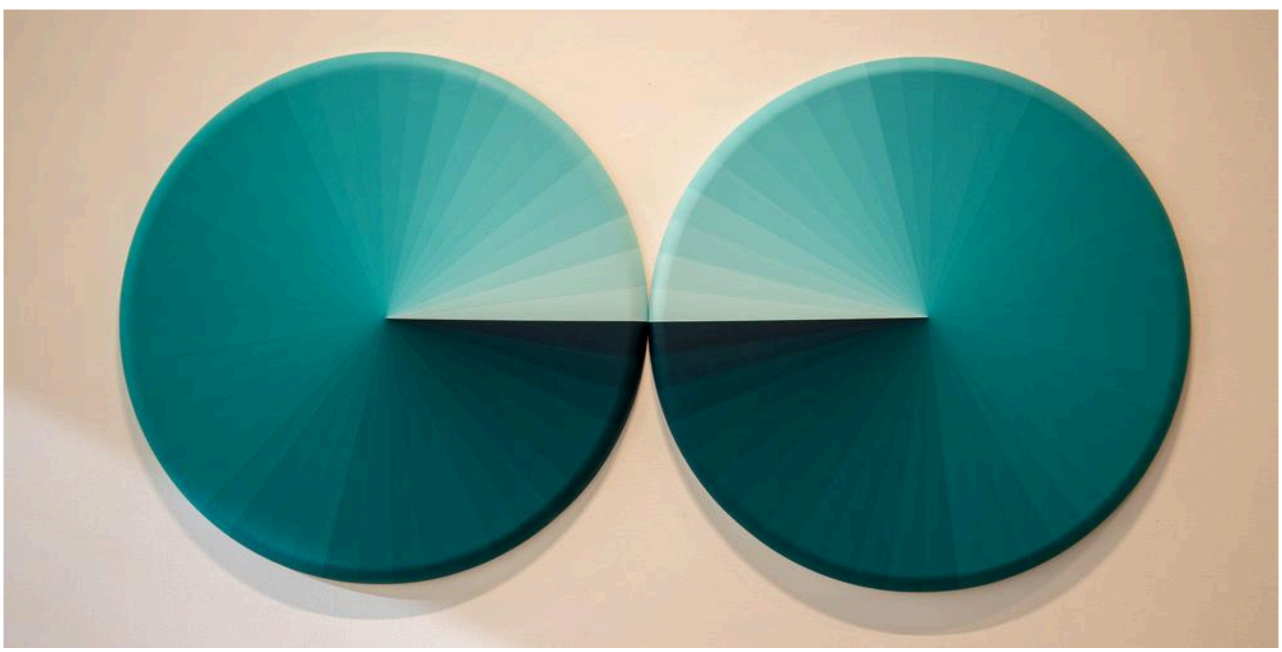

Figure 161: *Smack Dab in the Middle of Spectrum* by Jesse Stillwell, 2022. Acrylic on shaped canvas. 30"x60"

THE BRAIN IS AN INFERENCE MACHINE. The brain decides what we see by selecting the most plausible interpretation of visual information. Ernst Gombrich explores this idea in his landmark book *Art and Illusion*. His thesis is concisely illustrated by the duck/rabbit puzzle, also used by Ludwig Wittgenstein.

Figure 162: The duck-rabbit illusion appeared in Fliegende Blätter, a German magazine. It was later used by Ludwig Wittgenstein in *Philosophical Investigations*.

DO WE SEE A DUCK OR A RABBIT? Once you know that either answer is valid, you can see either animal. The visual information supplied by the same drawing can be mapped to internal representations of either object in the brain. Perception can wobble between duck or rabbit. You know that both animals can be seen, but you never see both animals at the same instant.

This illusion works because the drawing provides partial visual information about either animal. Neither animal is drawn photographically. The ambiguous drawing has enough duck-like and rabbit-like features that allow the brain to see either animal. Starting with partial information, your brain extrapolates the rest to see a duck or rabbit.

Every artwork, every man-made drawing has partial information. Because the brain unconsciously and automatically creates stable, coherent, and complete perception using partial visual information, a picture can look photographic even when it is not.

BILLY PAPPAS spent 8 years on one drawing to reach the limits of the pencil in achieving photographic accuracy. After applying 9,000,000 pencil marks to a sheet of paper (12.8"x16.3"), he drew his *Marilyn Monroe*. Even here, Pappas relies on illusion. Blonde hair cannot be drawn directly with a pencil. You see her hair only where the pencil does not make a mark. For 8 years, Pappas hung both arms from slings to keep them still in front of his easel. One hand held a magnifying glass. The other held the freshly sharpened pencil to make each stroke.

Figure 163: *Detail from Marilyn Monroe* by Billy Pappas, 2003.

Billy Pappas, *Marilyn Monroe*, 2003. Graphite on paper, 12.79 x 16.34 in. Courtesy of the artist. Digital image provided courtesy of William A. Christens-Barry, Chief Scientist, Equipoise Imaging, LLC.

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THE GIRL WITH A PEARL EARRING offers another kind of illusion. This painting is a *tronie* , a genre of portrait studies of facial expressions, especially those of Rembrandt and his followers. Vermeer owned *tronies* by two of Rembrandt's students, Samuel van Hoogstraten and Carel Fabritius. Vermeer himself painted several *tronies* – the *Girl with a Pearl Earring* is the best known. If the girl never existed, the painting is an illusion in entirety. If *The Girl* was painted from life, we do not know who she was. She might have been his daughter. But in 1665-1667, Vermeer's eldest daughter, Maria, was 11-13 years old.

Figure 164: Vermeer (Colin Firth) and Griet (Scarlett Johannssen) looking through a *camera obscura* in the film *Girl with a Pearl Earring*.

Lawrence Gowing, an artist and art historian, suggested that *The Girl with a Pearl Earring* was painted using a camera obscura. In the film adaptation of *The Girl with a Pearl Earring* – based on the historical novel by Tracy Chevalier – Scarlett Johanssen plays the part of the living model who looks through a camera obscura with Colin Firth, who plays Vermeer.

Figure 165: *The Girl with a Pearl Earring*

THE MAURITSHUIS MUSEUM collection has been entirely digitized, including *The Girl with a Pearl Earring*, at gigapixel resolution by Google Arts and Culture. The painting can be viewed at the ultra-high resolution afforded by Google's Art Camera.

Details emerge that reveal Vermeer's understanding of the art of illusion. For example, Vermeer does not paint the pearl by attempting to capture its photographic detail, point by point. Instead, Vermeer uses only two brushstrokes, one stroke to represent a reflection under another thick dab. Vermeer only supplies key details that are consistent with a large pearl. The mind of the viewer does the rest.

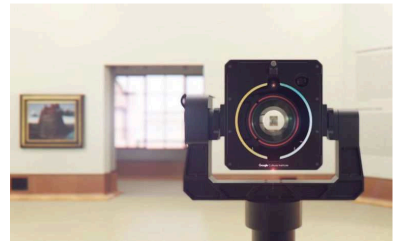


Figure 166: Google's Art Camera is deployed around the world to capture gigapixel images of masterpieces.

LOOK CAREFULLY AT THE BRIDGE OF HER NOSE. Vermeer paints over the bridge of the nose with the same hue as the cheek. The nose is an illusion, that you only see by inference based on its shadow.

Digital imaging and analysis reveals how Vermeer made his painting, layer by layer with painstaking detail. Abbie Vandivere at the Mauritshuis led the most recent technical analysis of *The Girl with a Pearl Earring* with non-invasive imaging tools with non-visible wavelengths. [10] Vermeer used multiple underlayers that varied in tone. He let underlayers dry before applying surface paint, using thinner or thicker upper layers in different regions of the painting. Paint thickness controlled how underlayers produced the appearance of light and shadow.

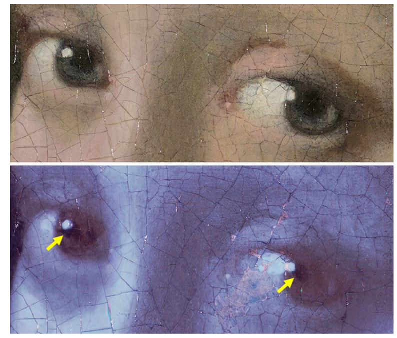


Figure 167: Evidence for pentimenti in the Girl's eyes. a Visible light photograph. b MS-IRR false colour detail. Dark marks indicate possible earlier iris locations (yellow arrows)

High-resolution imaging reveals other preparatory steps such as fine black outlines. Analysis also reveals *pentimenti*, subtle corrections and changes to the painting including relocations of the iris and ear. *The Girl With a Pearl Earring* was planned and perfected at many levels.

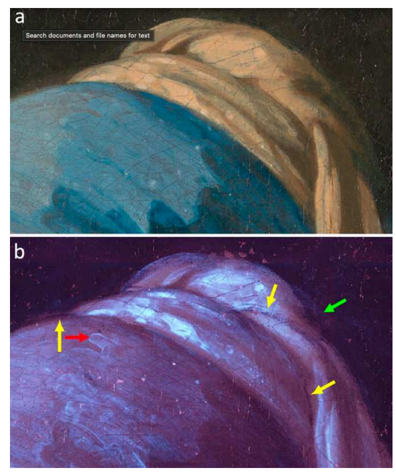
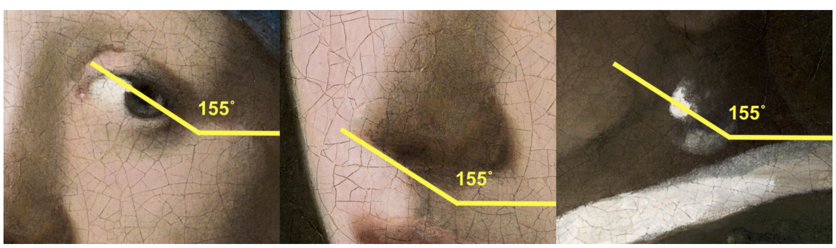


Figure 168: Evidence for contours and fine black outlines in the top of headscarf. a Visible light photograph. b MS-IRR false colour detail. Wavy brushstrokes at the surface (red arrow), fine black outlines applied in a preparatory phase (yellow arrows), back of the headscarf applied on top of an infrared-absorbing layer (green arrow).

Geometric analysis reveals remarkable consistency in the Vermeer's representation of incoming light that brightly reflects from the spheres of the eyes and pearls and that creates the shadow of the nose. All are $\sim 155^\circ$



[10] Abbie Vandivere, Annelies van Loon, Kathryn A. Dooley, Ralph Haswell, Robert G. Erdmann, Emilien Leonhardt, and John K. Delaney. Revealing the painterly technique beneath the surface of Vermeer's Girl with a Pearl Earring using macro- and microscale imaging. *Heritage Science*, 7:64, 2019

Writing

Consider any painting – one that you have seen so far in this class or any that you choose to consider – where the artist gives you an illusion of seeing something that isn't actually there.

Reading

Read the Introduction of *Art and Illusion* by Ernst Gombrich. Download "Introduction: Psychology and the Riddle of Style"

Bibliography

- Ernst Gombrich. *Art and Illusion: A Study in the Psychology of Pictorial Representation*. Princeton University Press, 2000. ISBN 0691070008
- David G Stork. *Pixels & paintings : foundations of computer-assisted connoisseurship*. Wiley, Hoboken, New Jersey, 2024. ISBN 9780470229446
- Abbie Vandivere, Annelies van Loon, Kathryn A. Dooley, Ralph Haswell, Robert G. Erdmann, Emilien Leonhardt, and John K. Delaney. Revealing the painterly technique beneath the surface of Vermeer's Girl with a Pearl Earring using macro- and microscale imaging. *Heritage Science*, 7:64, 2019

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Week Eight - Interaction of Color

Thursday, 27 March 2025, 12:45 - 2:15 PM EST.
Harvard Art Museums 0600

Figure 169: *A painter in his studio* Rembrandt Harmensz van Rijn (Anonymous pupil of) 1600-1700. Link to painting.

We see with four different types of photoreceptor cell. In dim light, we see with rod cells. In bright-light, we see with three cone cells that are tuned to different wavelengths of visible light. This is why rod vision is monochromatic and cone vision is trichromatic. Because the brain uses the activity of three different cone cells to decide what color it sees, any color that we see can be mixed from three different hues, such as red/green/blue or cyan/magenta/yellow.

Trichromatic vision is helpful to the painter, letting them create any visible color by mixing only a small number of pigments. The artist in Rembrandt's studio only appears to use seven pigments on his palette (Fig. 169). Leonardo da Vinci wrote that he only needed four different pigments – yellow, blue, green, and red – to create all colors. Yellow and blue make green. So da Vinci could have gotten away with three pigments.

Why do we see the specific wavelengths of visible light? Visible light happens to be inside a bandwidth of solar radiation that penetrates Earth's atmosphere. Daylight is evenly distributed across the wavelengths that we see, from 400 nm (blue) to 700 nm (red).

Newton discovered that white daylight could be separated into the spectrum of different colors using a prism (Fig. 170).

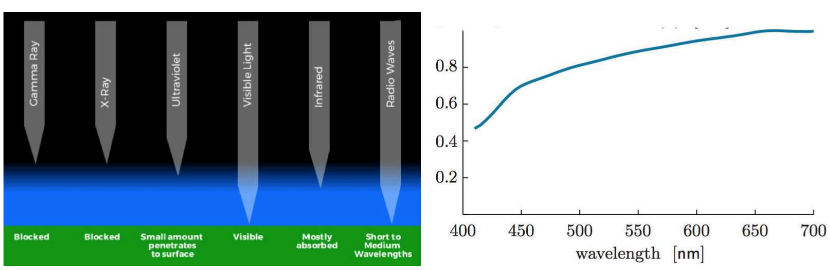

Figure 170: Most wavelengths of light are blocked by the atmosphere, except the visible bandwidth and X-rays. The latter is useful for X-ray telescopes. The spectrum of visible light is defined by a photon arrival rate function that is nearly constant over the range of visible light. From Nelson (2017)

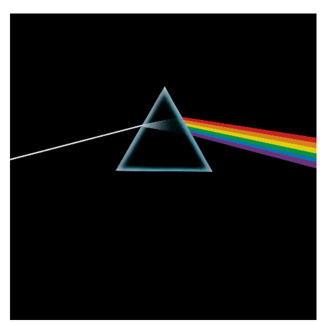

Figure 171: *Dark Side of the Moon*, Pink Floyd.

Cone cells are most densely packed in the fovea, the center of the field of view. The diameter of each cone cell is roughly the wavelength of visible light, a limit to spatial resolution. When you try to see spatial details that are smaller than your spatial resolution, the details will blur.

Our scientific understanding of the limits of spatial resolution came with the anatomical studies of Santiago Ramón y Cajal (1852-1934). Around the same time, George Seurat (1859-1891) and other pointillists were doing their own experiments involving the spatial resolution of color vision. Instead of brushstrokes, Seurat painted with small colored lights, individual dots and dashes of different color.

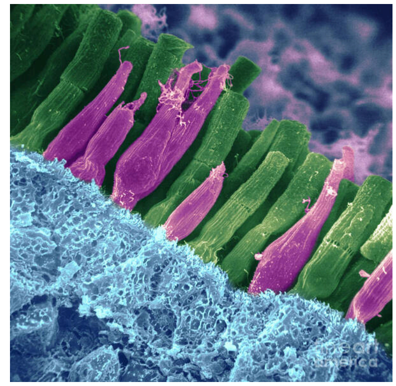

Figure 172: Art Print of pseudocolored electron micrograph of the retina, with green rod cells, red cone cells. From *www.pixels.com*.

Viewed up close Seurat's massive painting, *La Grande Jatte*, reveals his careful decision-making about the hue and tone of every dot. But viewed at a specific distance, the spatial resolution of the pointillism begins to match the spatial resolution of color vision. Just when cone cells start mixing the information from neighboring dots, the painting starts to shimmer. Pointillism is one step towards a new kind of art that directly strives for psychological sensation in the viewer. Seurat worked hard on this painting, Twenty-three preliminary drawings and thirty-eight painted color studies survive to testify to his struggle toward a desired effect.

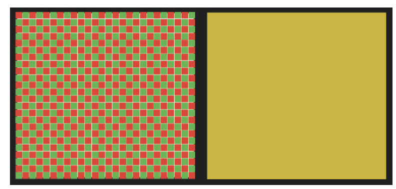

Figure 173: **Color illusion.** When viewed up close, the left box is seen to consist of small red and green squares. The image of each square on the back of the retina is larger than the size of photoreceptor cells. When viewed from afar, each photoreceptor receives light from both red and green squares, whose spectra merge. The perceived color becomes closer to the color in the right box than red or green. Red + Green ~ Yellow.

Figure 174: The juncture between dress, grass, and shadow reveal a variety of color choices, no single tone or hue.

Figure 175: *A Sunday on La Grande Jatte* George Seurat, 1884-1886. His best known painting is also his largest, 81.7 in x 121 x in. Art Institute of Chicago.

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However, we do not see color on the basis of pixels. Information from individual cone cells is not sent directly to the brain. Instead, the retina integrates over areas of color information, comparing and contrasting the information from adjacent fields to decide what color to assign to each field. The brain uses color differences at boundaries to assign color, analogous to how it uses sudden differences in brightness to detect edges. Color depends on context.

Josef Albers, 1888-1976, was a German artist and educator who wrote the seminal work on color theory of artists, *The Interaction of Color.* Here, I will select a few of his lessons.

First, the same color has many faces. Color is relative.

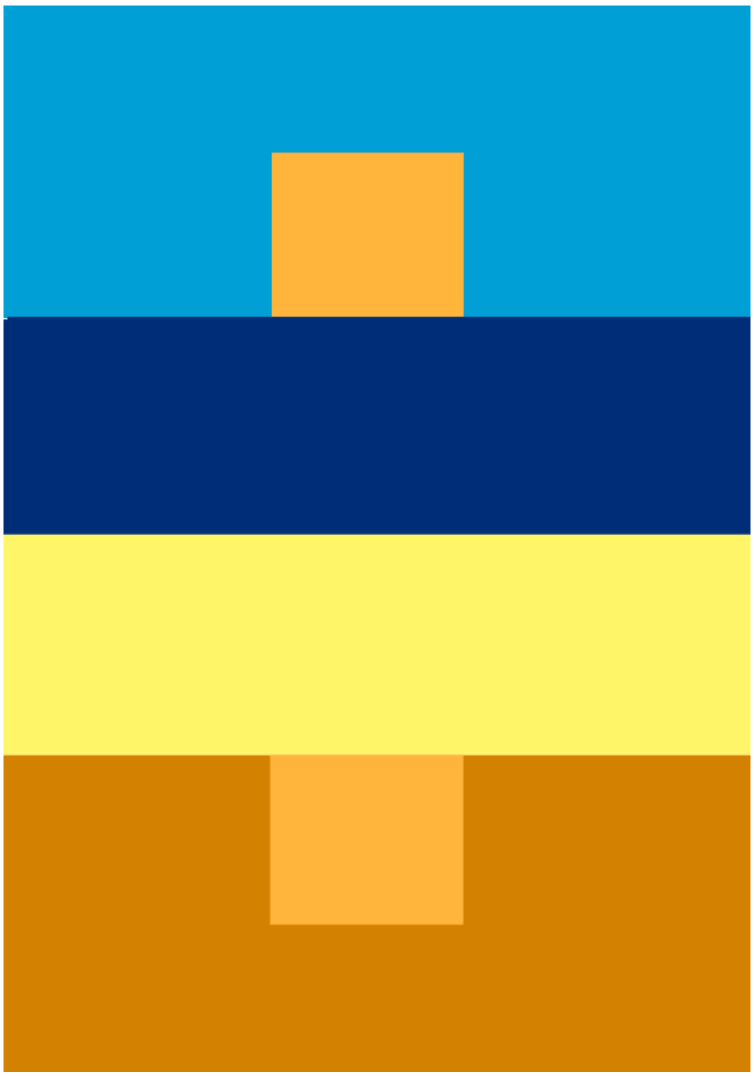

Figure 176: Plate IV-1. The upper small and the lower small squares are the same color, although the upper one looks more 'orange' and the lower one looks more 'tan'.

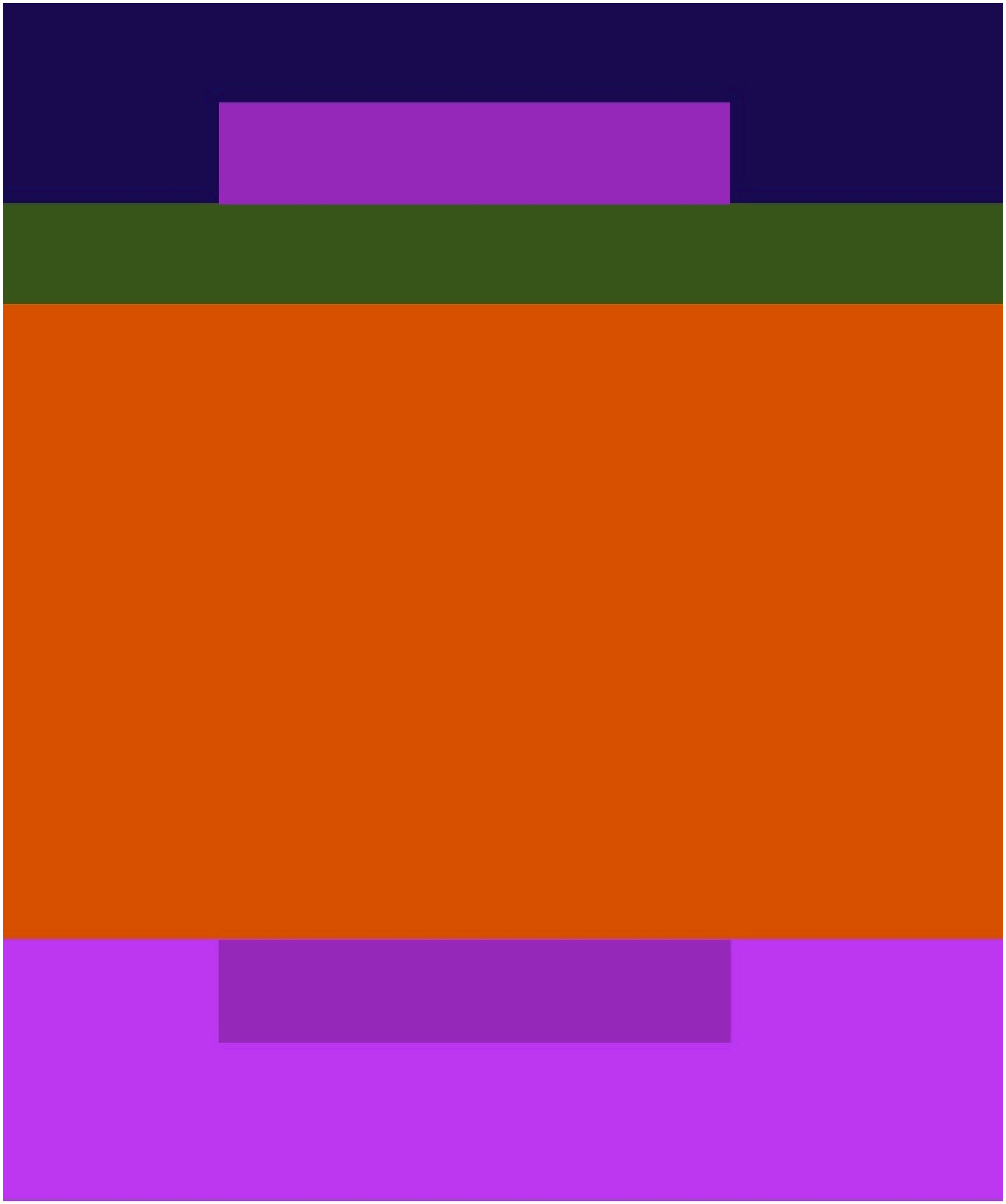

Figure 177: Plate IV-4. The inner smaller violets are the same, but the upper one looks more like the outer violet at the bottom.

AFTER-IMAGES are another way to experience how the eye determines color. Stare at the red circle for 30 s, keeping your eye on the black center. Then look at the black center of the white circle. It does not appear white. While you stare at the red circle, the retina is adapting. The photoreceptors for red light are strongly activated, and gradually begin to inhibit the output of neighboring red photoreceptors. This inhibition is long-lasting. So when you look at the white circle, the retina is effectively subtracting the red light from the white light, leaving blue/green light.

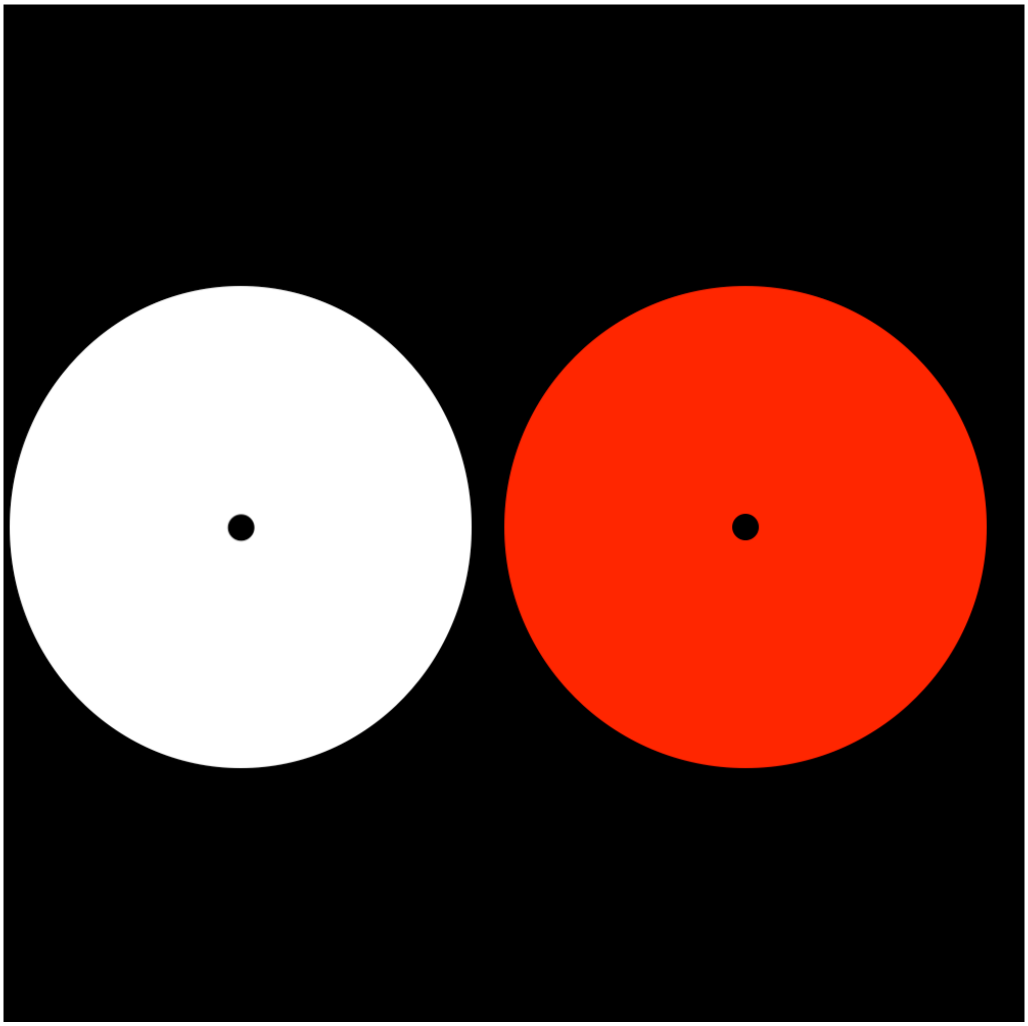

Figure 178: Plate VIII-I.

DOUBLE-REVERSAL is seen in this experiment. After staring at the set of yellow circles in the white box, the eye gradually adapts by inhibiting yellow perception inside the circles. When you shift your eye to the white box, yellow diamonds appear.

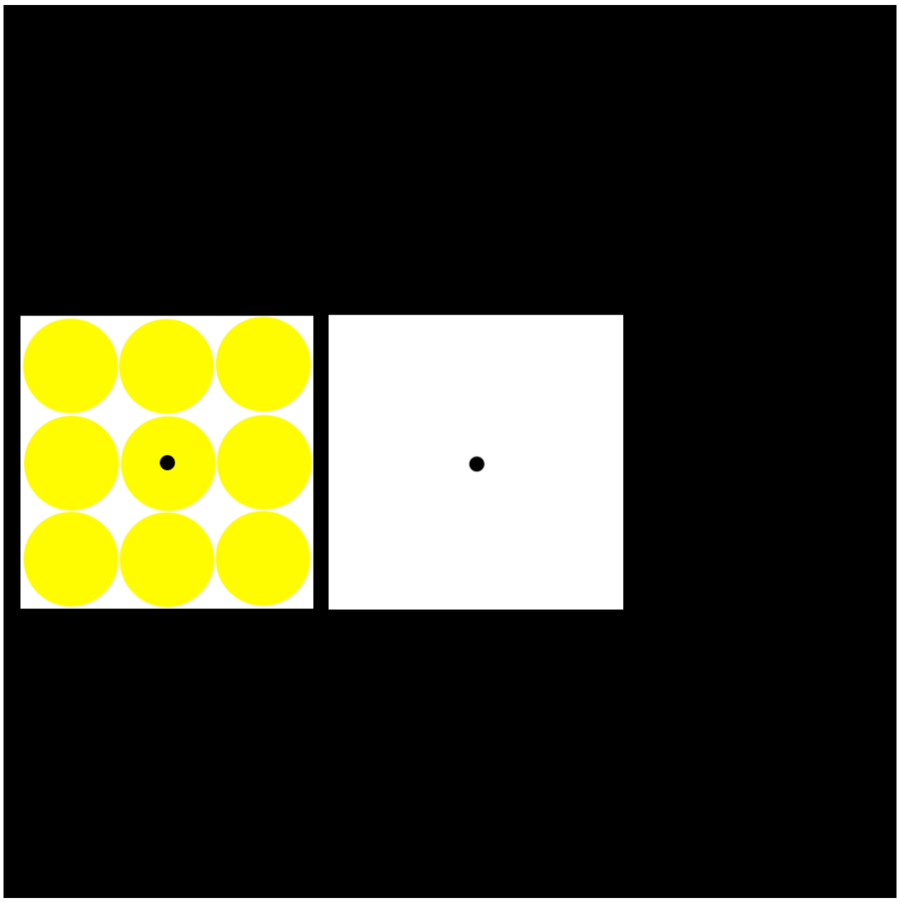

Figure 179: Plate VIII-2.

THE COLOR-OPPONENCY THEORY came from the systematic analysis of 'after-image' effects using different colors, and from hue-cancellation effects where one is asked what colored light will cancel a test color. Color-opponency theory is the basis of the color wheel that is widely used by artists.

While color-coding in the retina might be relatively straightforward – based on three cone types and inhibitory and cross-inhibitory circuitry – there is no simple algebra that converts retinal information into a reliable prediction of how each person sees color in any picture. Each human brain takes the whole field-of-view into account when deciding colors. Color-opponency theory – derived from simple experiments and color arrangements – does not fully explain color perception in art of life. A viral meme from 2015 (#thedress) makes this point.

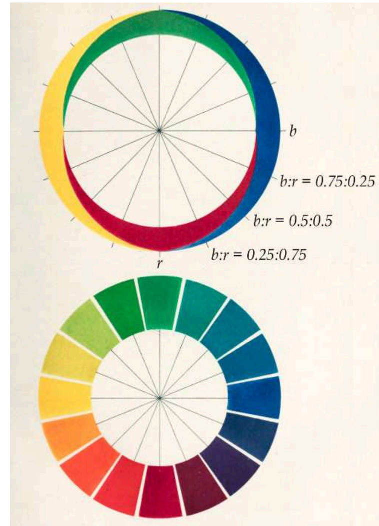

Figure 180: Hering's Color Opponency Theory, from Conway et al. (2023).

CLAUDE MONET (1840-1926) uses color-opponency when he uses blue to paint the shadows of haystacks in yellowing twilight.

Figure 182: *Wheatstacks, Snow Effect, Morning* by Claude Monet, 1891. Getty Museum.

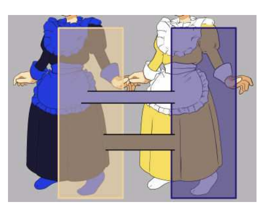

Figure 181: When this photo of a dress was posted to Facebook in 2015, it went viral. Many saw a blue and black dress. Many saw a white and gold dress. The answer depends on how you interpret the illuminating light, whether you see a blue and black dress that is illuminated by yellowish light or a white and gold dress that is illuminated by bluish light.

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COLOR VISION assigns the hue of an object, but not identify the edges of an object. We use brightness information – spatial differences in luminosity – to find object boundaries. Josef Albers makes this clear with overlapping bars of different color and brightness. The lower horizontal bar appears to overlap the center bar, whereas the upper horizontal bar is overlapped by the center bar. The middle bar is ambiguous. The brain draws outlines around adjacent visual fields, connecting fields into single objects when similar contrasts in brightness occur at a shared outer edge.

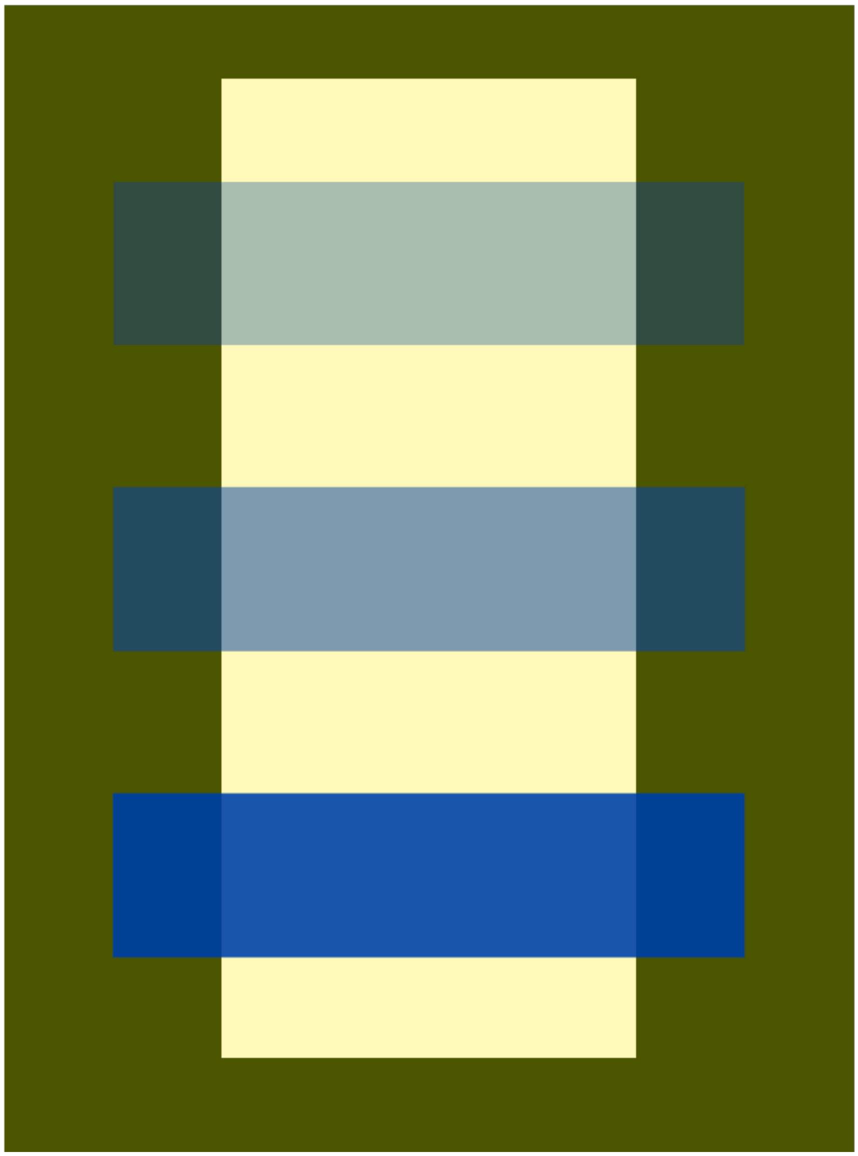


Figure 183: Plate XI-3.

IMPRESSION, SUNRISE was painted by Monet in 1872. The name of the impressionist movement came from this painting of the port of Le Havre. The haziness and loose brush strokes broke from traditional landscape painting. Objects are hardly distinguishable and boundaries seem to vanish. Edge detection based on sudden changes in brightness is subverted. The sun shimmers. The sun lacks a well-defined boundary in the cloudy sky.

HOW DOES MONET CREATE THE ILLUSION of the shimmering sun? He used color mixes for sun and sky with equal brightness. Because the brain uses brightness, not color, to draw object outlines, it has trouble finding the edges of the sun. And so the sun appears to pulse and vibrate in the sky. This is made clear by shifting the painting to grayscale – the sun disappears.

PLUS REVERSED, oil on canvas, by Richard Anuszkiewciz (1930-2020) is a conscious effort toward pulsation and vibration by using isoluminous regions of red and green. The effect can be unsettling and is usually avoided by graphic designers. But an artist might want to unsettle their viewer.

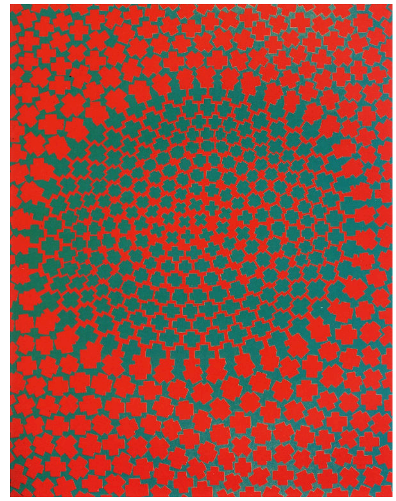


Figure 184: *Plus Reversed Sunrise* by Richard Anuszkiewciz, 1960. The equally luminous green and red fields shimmer and dance.

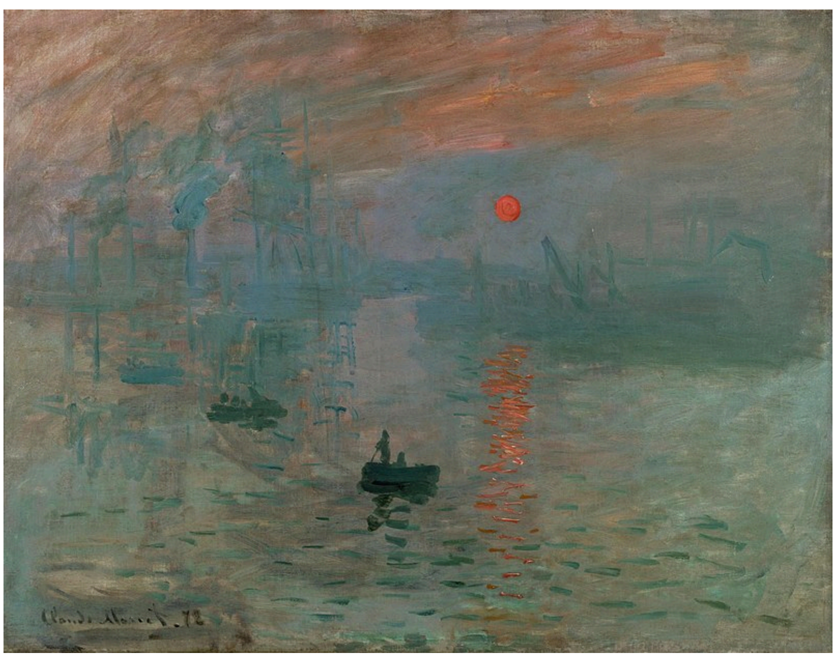
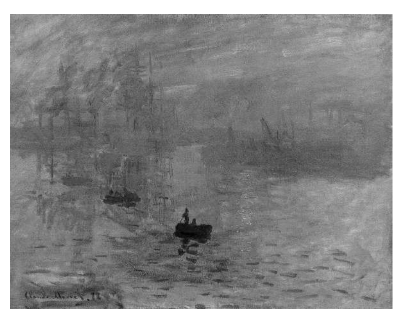


Figure 185: *Impression, Sunrise* by Claude Monet. Below, the same painting in grayscale. Almost all luminosity boundaries disappear, including between sun and the sky. By subverting your brain's ability to see object boundaries using luminance, the brain is confused.



Ochre and pink are easily recognized as different colors when presented separately. But when equal in luminosity, it becomes difficult to see the boundary between these colors when they are adjacent.

Figure 186: Plate XXIII-1

André Derain uses a range of different colors when painting the face of Henri Matisse. But by using equal luminosity in the many-colored brush strokes, the face keeps its and is unfragmented at color boundaries.

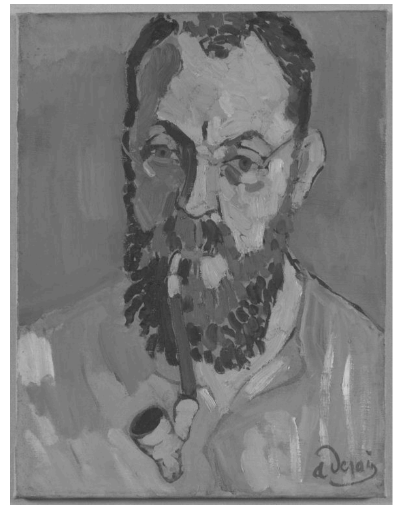


Figure 187: *Portrait of Henry Matisse* by André Derain, 1905. Tate Gallery.

IMPRESSIONISM broke from Western tradition in many ways. From the Renaissance to the 19th century, art was largely narrative. Pictures told stories or faithfully transcribed what was seen. Color was largely decorative, essentially filling in what could have been (or was) drawn.

The impressionists used color to compose the image. *Impression, Sunrise* and *La Grande Jatte* are painted with color and light. Without color, their images dissolve. In this way, Monet and Seurat released visual art from prior assumptions.

PAUL CEZANNE (1839-1906) bridged impressionism and modern art by finding new ways to release art from prior assumptions. The Dutch still lifes of Pieter Claesz or Willem Kalf were displays of virtuosity and optical precision. Cezanne painted many still lifes, where depth is modeled with color, not with linear perspective or shading. Nature is not intentionally distorted, but distortion was acceptable if the still life was no less convincing.

Figure 188: *Still Life with Commode* by Paul Cezanne, 1887-1888. Harvard Art Museums.

MODERN ART has made total break from narrative purpose. Mark Rothko (1903-1970) essentially painted nothing but color and light. A 'Rothko' means a canvas filled with floating blocks of color. Painting pure color, Rothko was strict about the lighting and placement of his canvases.

Figure 189: *No. 3/No. 13* by Mark Rothko, 1949. MoMA.

In 1962, Rothko painted ten murals for a dining room on the top floor of Holyoke Center at Harvard. The murals were displayed until 1970s when they were had to be removed because of damage and fading. Color photographs reveal what they originally looked like. In 2014, these paintings were brought out of storage and 'digitally restored' for a special exhibition at the Harvard Art Museums, using a projector to precisely compensate for their fading.

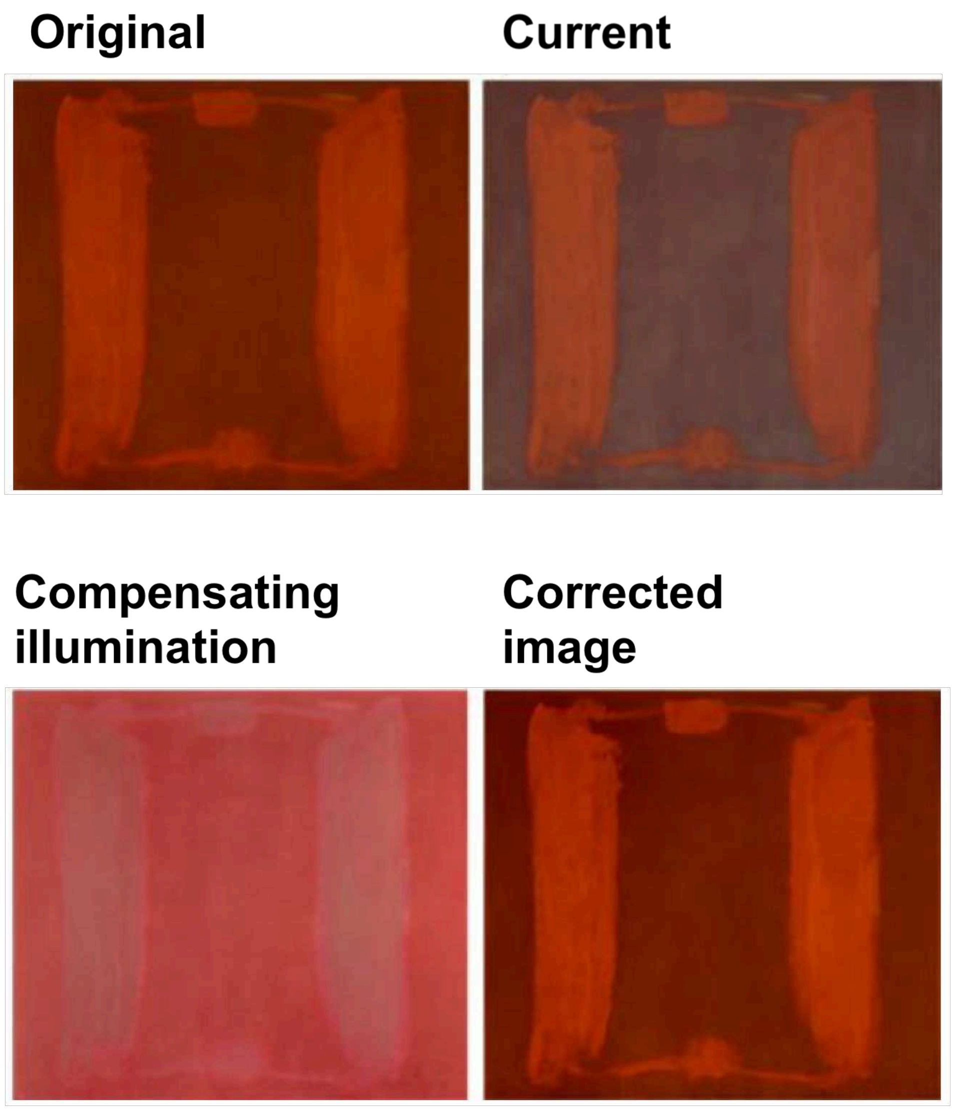


Figure 190: *Panel One* by Mark Rothko, 1962 and digital restoration.